rogoni.market Continuous Learning Architecture

Version 1.0

Purpose

This document defines a continual learning system for Grogoni and the Grogoniclaw runtime. Rather than relying on model fine-tuning, the platform continuously improves through runtime, skills, memory, evaluation, and production feedback.

Core Principle

The intelligence of Grogoni lives primarily in the runtime rather than the language model.

Every customer interaction, business operation, tool execution, and workflow becomes data that can improve the platform. Improvements are implemented in the planner, router, prompts, skills, memory, verification, and execution engine, allowing all agents to benefit.

Learning Layers

1. Model Layer – interchangeable LLMs.
2. Harness Layer – planner, router, OGIV, tool selection, prompts, retries, verification.
3. Context Layer – business, customer, supplier, inventory and market memories.
4. Skill Layer – reusable workflows extracted from successful executions.
5. Ecosystem Layer – anonymized learning shared across participating stores.

Continuous Learning Pipeline

User Request
→ Execution Graph
→ Trace Capture
→ OGIV Validation
→ Memory Snapshot
→ Outcome Classification (Success / Failure / Partial Success)
→ Pattern Discovery
→ Skill Candidate Detection
→ Evaluation Engine
→ Runtime Improvement Proposal
→ Human Review
→ Deployment
→ Every Store Benefits

Execution Trace

Capture prompts, plans, tool calls, latency, errors, business outcomes, user corrections, purchases and task completion. Store traces with tenant isolation.

Memory System

Maintain customer memory, shop memory, supplier memory, market memory and runtime memory. Promote frequently reused knowledge into reusable skills.

Skill Promotion

When a workflow repeatedly succeeds, convert it into a versioned skill package with tests and metadata. Skills may be private, organisation-wide or marketplace assets.

rogoniBench

Introduce an evaluation benchmark for commerce. Example tasks include creating stores, processing orders, inventory updates, supplier onboarding, checkout, delivery and customer support. Every runtime change must pass GrogoniBench before release.

Production Intelligence

Cluster production failures and successes, identify recurring patterns, prioritise improvements and generate candidate fixes. Human engineers approve production deployments.

Roadmap

Phase 1: Trace collection.
Phase 2: Unified memory engine.
Phase 3: Automatic skill extraction.
Phase 4: GrogoniBench.
Phase 5: Runtime self-improvement assistant.
Phase 6: Marketplace-wide learning.

Expected Outcomes

- Better agent performance without retraining models.
- Faster deployment of improvements.
- Shared learning across stores.
- Higher task completion.
- Lower operational cost.
- Continuously improving commerce agents.